

Sales Data Analysis Project Using Python and Power BI

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Chapter 1

Project Overview

This project aims to analyze sales data to extract meaningful insights about performance, profitability, and regional distribution. The analysis combines the power of **Python** for data processing and **Power BI** for interactive visualization.

Objectives

- Clean and preprocess raw sales data.
- Perform exploratory data analysis (EDA) to identify trends and correlations.
- Build visual dashboards in Power BI for decision-making.
- Apply basic predictive modeling to forecast profit.

Chapter 2

Data Sources

- **Dataset:** Superstore Sales Dataset.
- **Format:** CSV file.
- **Columns:** Order Date, Sales, Profit, Quantity, Category, Region, Segment, Product Name, etc.
- **Data Source:** Kaggle / Internal Database.

Chapter 3

Data Collection and Loading

The dataset was imported into Python using the Pandas library:

```
import pandas as pd
data = pd.read_csv("superstore_sales.csv")
```

Chapter 4

Data Cleaning and Transformation

The following steps were performed:

- Removed missing and duplicate records.
- Converted date columns to datetime format.
- Created derived fields such as Year, Month, and Profit Margin.
- Handled inconsistent data types and outliers.

Chapter 5

Exploratory Data Analysis (EDA)

Descriptive and visual analysis was conducted to understand the dataset.

Key Steps

- Computed summary statistics using `data.describe()`.
- Generated correlation matrices using Seaborn heatmaps.
- Visualized sales trends over time.
- Compared category-wise and regional performance.

Example Code

```
import seaborn as sns
import matplotlib.pyplot as plt

sns.heatmap(data.corr(), annot=True, cmap='coolwarm')
plt.show()
```

Chapter 6

Advanced Analysis

A simple Linear Regression model was used to predict profit based on sales and quantity.

Code Example

```
from sklearn.linear_model import LinearRegression

X = data[['Sales', 'Quantity']]
y = data['Profit']

model = LinearRegression()
model.fit(X, y)
y_pred = model.predict(X)
```

Evaluation

Performance metrics such as R^2 score and RMSE were calculated to assess accuracy.

Chapter 7

Power BI Dashboard

The cleaned dataset was imported into Power BI to create an interactive dashboard.

Dashboard Pages

1. **Sales Trends:** Line chart showing sales over time.
2. **Product Performance:** Bar chart of top-selling products.
3. **Regional Sales:** Map showing sales by region.
4. **Customer Segments:** Pie chart for sales by customer type.

Interactivity

- Filters for Year, Category, and Region.
- Slicers to dynamically update charts.

Chapter 8

Key Insights

- Sales peak in Q4 every year, mainly in the Technology category.
- West region generates the highest profit margin.
- Office Supplies have high sales but low profit.
- Strong positive correlation between Sales and Profit (0.75).

Chapter 9

Tools and Technologies

Tool	Purpose
Python (Pandas, Matplotlib, Seaborn)	Data Cleaning and EDA
Scikit-learn	Predictive Modeling
Power BI	Data Visualization
Excel / CSV	Data Storage

Chapter 10

How to Use the Power BI Dashboard

1. Open the `Superstore_Dashboard.pbix` file in Power BI Desktop.
2. Use slicers to filter by year, category, or region.
3. Hover over visuals to see detailed tooltips.
4. Export or publish the report to Power BI Service.

Chapter 11

Conclusion

This project provided valuable insights into sales patterns and profit distribution. The combination of Python and Power BI created a full analytical workflow — from raw data to dashboards.

Future Work

- Develop predictive models for future sales.